Course Outline

Department of Mathematics and Physical Sciences
BRAC University

MAT 120: Integral Calculus and Differential Equations

A. Course General Information

Course Code:	MAT 120
Course Title:	Integral Calculus and Differential Equations
Credit Hours:	3
Contact Hours/Week:	Two lectures (1h20m each) + Laboratory
Category:	Offered by MPS for other programs
Type:	Theory + Laboratory
Prerequisites:	MAT 110
Co-requisites:	None
Course Instructor:	
Consultation Hours:	

B. Course Catalog Description

Integral Calculus: Integration using Riemann sums; antiderivatives; indefinite and definite integrals; substitution; improper integrals; integration by parts; reduction formula; trigonometric substitution; integration of rational functions; Beta and Gamma functions; arc length; surface area of revolution; double integrals (Cartesian & polar); triple integrals (Cartesian, cylindrical & spherical); change of variables and the Jacobian.

Differential Equations: Introduction; separable and linear equations; integrating factor; exact and non-exact (made exact); higher-order homogeneous & non-homogeneous ODEs; modeling with first-order ODEs.

C. Rationale of the Course

Integral Calculus and Differential Equations form a core mathematical foundation for engineering and applied sciences, enabling modeling and solution of diverse real-world problems and preparing students for advanced coursework.

D. Course Objectives

- Develop a clear understanding of integral calculus and differential equations.
- Practice writing and proving results in an organized mathematical style.
- Apply techniques to efficiently solve nontrivial problems.
- Strengthen critical thinking as future engineers via applications of integrals and ODEs.

E. Course Outcomes (COs)

SL.	CO Description
CO1	Comprehend integration for single and multivariable functions.
CO2	Build models in several branches of engineering using integral calculus.
CO3	Understand the theory of differential equations.
CO4	Apply differential equations to solve real-world problems.

F. Course Plan (Weekly Schedule)

Week	Topics	Teaching–Learning Strategy	Assessment Strategy	COs
1	Definition of integration; antiderivatives.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2
2	Techniques for definite/indefinite integrals: u-substitution, trig. substitution, parts, reduction formulas, rational functions.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2
3	Improper integrals; arc length; surface area of revolution.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2
4	Beta and Gamma functions.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2
Midterm Coverage: Weeks 1–4 (tentative)				
5	Double integrals: rectangular and polar coordinates.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2
6	Triple integrals: rectangular, cylindrical, spherical.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2
7	Change of variables; Jacobian.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO1, CO2

Week	Topics	Teaching–Learning Strategy	Assessment Strategy	COs
8	Introduction to differential equations.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO3, CO4
9	First-order ODEs: separable and linear (integrating factor).	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO3, CO4
10	Exact and non-exact equations (made exact).	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO3, CO4
11	Higher-order ODEs: homogeneous and non-homogeneous.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO3, CO4
12	Modeling with first-order ODEs.	Lecture, discussion, consultation.	Assignment, quiz, examinations.	CO3, CO4
Final Coverage: Weeks 5–12 (tentative)				

G. Learning Materials

References:

- H. Anton, *Calculus: Early Transcendentals* (10th ed.).
- D. G. Zill, *A First Course in Differential Equations with Modeling Applications* (7th ed.).

H. Delivery Method

Two lectures per week (each 1 hour 20 minutes). PowerPoint and other teaching aids as needed. Classroom discussion and problem-solving sessions supplement lectures; laboratory activities support applications.

I. Assessment and Evaluation

Component	Attendance	Assignment	Quiz	Lab	Midterm	Final
Weight (%)	5	10	15	10	20	40

Attendance Policy: Arriving 15 minutes late or more is recorded as absent. Students with total attendance (lecture + lab) below 70% are not permitted to sit for midterm and final exams and will be listed as absentee.

Attendance (%)	Marks
90 and above	5
85–89	4
80–84	3
75–79	2
70–74	1
Less than 70	0 (Absentee)

Make-up Procedure: Make-ups are only for extreme medical conditions or death in the immediate family, recommended by the course teacher and approved by the department head; standard make-up exam fees apply.

J. Grading Policy

% of Marks	Letter Grade	Grade Points
97–100	A+	4.00
90–<97	A	4.00
85–<90	A-	3.70
80–<85	B+	3.30
75–<80	B	3.00
70–<75	B-	2.70
65–<70	C+	2.30
60–<65	C	2.00
57–<60	C-	1.70
55–<57	D+	1.30
52–<55	D	1.00
50–<52	D-	0.70
<50	F	0.00